

Competency Questions: DL Query and SPARQL Formalisation

This document lists all competency questions used in the road-sign ontology project and gives the most appropriate formalisation for each one. Questions that can be answered as class expressions are translated as Protégé DL queries. Questions about property metadata, equivalence mappings, labels, image metadata, or annotation values are translated as SPARQL queries.

DL Query convention: run the query in Protégé with the indicated result mode, usually *Subclasses*, *Superclasses*, or *Direct subclasses*.

SPARQL convention: use the standard RDF/RDFS/OWL prefixes below. Replace the ontology namespace in the first prefix if your ontology IRI is different.

```
PREFIX : <http://example.org/vienna-road-signs#>
PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
```

1. CQs about the taxonomy of road signs

Competency question: What are the main categories of road signs in the Vienna Convention?

Method: DL Query

Protégé option: Direct subclasses

Translation:

RoadSign

Note: Retrieves the direct subclasses of RoadSign.

Competency question: Is a danger warning sign a type of road sign?

Method: DL Query

Protégé option: Superclasses

Translation:

DangerWarningSign

Note: RoadSign should appear as a superclass.

Competency question: Is a prohibitory sign a type of regulatory sign?

Method: DL Query

Protégé option: Superclasses

Translation:

ProhibitorySign

Note: If the ontology uses the broader class name, also test ProhibitoryOrRestrictiveSign.

Competency question: Is a mandatory sign a type of regulatory sign?

Method: DL Query

Protégé option: Superclasses

Translation:

MandatorySign

Note: RegulatorySign should appear as a superclass.

Competency question: Which classes are subclasses of RoadSign?

Method: DL Query

Protégé option: Subclasses

Translation:

RoadSign

Note: Retrieves all classes below RoadSign, not only the direct categories.

Competency question: Which signs belong to the category of additional panels?

Method: DL Query

Protégé option: Subclasses

Translation:

AdditionalPanel

Note: Retrieves the additional-panel branch.

Competency question: Which signs are informative rather than regulatory?

Method: DL Query

Protégé option: Subclasses

Translation:

InformativeSign

Note: Retrieves the informative-sign branch.

2. CQs about road-sign properties

Competency question: What shapes are associated with each traffic sign category?

Method: DL Query + Python recommended

Protégé option: Subclasses

Translation:

`RoadSign and (hasShape some SignShape)`

Note: DL Query retrieves signs with shape restrictions. The exact shape per category is better listed with the Owlready2 script.

Competency question: What is the numerical speed limit value represented by a speed limit sign?

Method: DL Query

Protégé option: Subclasses

Translation:

`RoadSign and (hasMaximumSpeedValue some xsd:decimal)`

Note: If Protégé does not parse xsd:decimal, evaluate the data property with SPARQL.

Competency question: What is the minimum speed value represented by a compulsory minimum speed sign?

Method: DL Query

Protégé option: Subclasses

Translation:

`RoadSign and (hasMinimumSpeedValue some xsd:decimal)`

Competency question: What is the advisory speed value represented by an advisory speed sign?

Method: DL Query

Protégé option: Subclasses

Translation:

`RoadSign and (hasAdvisorySpeedValue some xsd:decimal)`

Competency question: What is the maximum permitted height represented by a height restriction sign?

Method: DL Query

Protégé option: Subclasses

Translation:

`RoadSign and (hasMaximumHeightValue some xsd:decimal)`

Competency question: What is the maximum permitted width represented by a width restriction sign?

Method: DL Query

Protégé option: Subclasses

Translation:

`RoadSign and (hasMaximumWidthValue some xsd:decimal)`

Competency question: What is the maximum permitted length represented by a length restriction sign?

Method: DL Query

Protégé option: Subclasses

Translation:

`RoadSign and (hasMaximumLengthValue some xsd:decimal)`

Competency question: What is the maximum permitted laden mass represented by a weight restriction sign?

Method: DL Query

Protégé option: Subclasses

Translation:

`RoadSign and (hasMaximumWeightValue some xsd:decimal)`

Competency question: What is the maximum permitted axle mass represented by an axle-weight restriction sign?

Method: DL Query

Protégé option: Subclasses

Translation:

`RoadSign and (hasMaximumAxleWeightValue some xsd:decimal)`

Competency question: What distance value is displayed on an additional panel?

Method: DL Query

Protégé option: Subclasses

Translation:

`RoadSign and (hasDistanceValue some xsd:decimal)`

Competency question: What textual information is displayed on an information sign?

Method: DL Query

Protégé option: Subclasses

Translation:

`RoadSign and (hasTextualInformation some xsd:string)`

Note: If datatype parsing fails, use SPARQL over datatype properties.

Competency question: What destination text is displayed on a direction sign?

Method: DL Query

Protégé option: Subclasses

Translation:

`RoadSign and (hasDestinationText some xsd:string)`

Competency question: What service name is displayed on a service sign?

Method: DL Query

Protégé option: Subclasses

Translation:

`RoadSign and (hasServiceName some xsd:string)`

Competency question: What road-sign code is associated with a sign class?

Method: SPARQL

Translation:

```
SELECT ?property ?domain ?range
WHERE {
  ?property rdf:type owl:DatatypeProperty .
  ?property rdfs:domain ?domain .
  ?property rdfs:range ?range .
  FILTER(CONTAINS(STR(?property), "hasSignCode"))
}
```

Note: This is a literal-valued data property, so SPARQL is clearer than DL Query.

Competency question: Which properties are functional?

Method: SPARQL

Translation:

```
SELECT ?property
WHERE {
  ?property rdf:type owl:FunctionalProperty .
}
ORDER BY ?property
```

Note: FunctionalProperty is property metadata, not a class expression.

Competency question: Which object properties are inverse properties of each other?

Method: SPARQL

Translation:

```
SELECT ?property ?inverseProperty
WHERE {
  ?property owl:inverseOf ?inverseProperty .
}
ORDER BY ?property
```

Note: Inverse properties are property axioms, so SPARQL is the appropriate formalisation.

3. CQs about the pragmatics of road signs

Competency question: Which type of road user does a sign apply to?

Method: DL Query

Protégé option: Subclasses

Translation:

```
RoadSign and (appliesToRoadUser some RoadUser)
```

Note: For a specific road user, use e.g. Pedestrian.

Competency question: Which type of vehicle does a regulatory sign apply to?

Method: DL Query

Protégé option: Subclasses

Translation:

```
RegulatorySign and (appliesToVehicle some Vehicle)
```

Competency question: What regulation is expressed by a regulatory sign?

Method: DL Query

Protégé option: Subclasses

Translation:

```
RoadSign and (expressesRegulation some TrafficRegulation)
```

Competency question: What prohibition is expressed by a prohibitory sign?

Method: DL Query

Protégé option: Subclasses

Translation:

```
RoadSign and (expressesProhibition some Prohibition)
```

Competency question: What obligation is expressed by a mandatory sign?

Method: DL Query

Protégé option: Subclasses

Translation:

```
RoadSign and (expressesObligation some Obligation)
```

Competency question: What restriction is expressed by a restriction sign?

Method: DL Query

Protégé option: Subclasses

Translation:

RoadSign and (expressesRestriction some Restriction)

Competency question: What hazard is indicated by a warning sign?

Method: DL Query

Protégé option: Subclasses

Translation:

RoadSign and (warnsAbout some RoadHazard)

Competency question: What road condition is indicated by a warning sign?

Method: DL Query

Protégé option: Subclasses

Translation:

RoadSign and (indicatesRoadCondition some RoadCondition)

Competency question: What priority rule is expressed by a priority sign?

Method: DL Query

Protégé option: Subclasses

Translation:

RoadSign and (indicatesPriorityRule some PriorityRule)

Competency question: What direction is indicated by a direction sign?

Method: DL Query

Protégé option: Subclasses

Translation:

RoadSign and (indicatesDirection some Direction)

Competency question: What destination is indicated by a direction sign?

Method: DL Query

Protégé option: Subclasses

Translation:

RoadSign and (indicatesDestination some Destination)

Competency question: What service is indicated by a service sign?

Method: DL Query

Protégé option: Subclasses

Translation:

RoadSign and (indicatesService some Service)

Competency question: Which additional panel modifies a main road sign?

Method: DL Query

Protégé option: Subclasses

Translation:

RoadSign and (modifiesSign some RoadSign)

Competency question: Which main sign is modified by an additional panel?

Method: DL Query

Protégé option: Subclasses

Translation:

`RoadSign and (isModifiedByPanel some AdditionalPanel)`

Note: This requires restrictions using `isModifiedByPanel`; otherwise use the inverse property axiom in SPARQL.

Competency question: Can the ontology distinguish between signs expressing prohibitions, obligations, restrictions, hazards, directions, services, and priority rules?

Method: DL Query group

Protégé option: Subclasses

Translation:

`RoadSign and (expressesProhibition some Prohibition)`
`RoadSign and (expressesObligation some Obligation)`
`RoadSign and (expressesRestriction some Restriction)`
`RoadSign and (warnsAbout some RoadHazard)`
`RoadSign and (indicatesDirection some Direction)`
`RoadSign and (indicatesService some Service)`
`RoadSign and (indicatesPriorityRule some PriorityRule)`

Note: Run the group of queries and compare the returned classes.

4. CQs relevant to the particular dataset: MTDS

Competency question: Which MTDS classes are represented in the ontology?

Method: DL Query

Protégé option: Subclasses

Translation:

`MTDSDatasetSignClass`

Competency question: Which MTDS classes are subclasses of existing road-sign categories?

Method: DL Query

Protégé option: Subclasses

Translation:

`MTDSDatasetSignClass and RoadSign`

Competency question: Which MTDS classes are equivalent to Vienna Convention road-sign classes?

Method: SPARQL

Translation:

```
SELECT ?class ?equivalentClass
WHERE {
  ?class owl:equivalentClass ?equivalentClass .
  FILTER(CONTAINS(STR(?class), "MTDS") || CONTAINS(STR(?equivalentClass), "MTDS"))
}
ORDER BY ?class
```

Note: Equivalence mappings are OWL axioms and are best retrieved with SPARQL.

Competency question: Are all MTDS classes connected to the RoadSign hierarchy?

Method: DL Query

Protégé option: Subclasses

Translation:

`MTDSDatasetSignClass and RoadSign`

Note: Compare the result with all subclasses of `MTDSDatasetSignClass`.

Competency question: Which MTDS classes belong to warning signs?

Method: DL Query

Protégé option: Subclasses

Translation:

MTDSWarningSignClass

Competency question: Which MTDS classes belong to regulatory signs?

Method: DL Query

Protégé option: Subclasses

Translation:

MTDSRegulatorySignClass

Competency question: Which MTDS classes belong to mandatory signs?

Method: DL Query

Protégé option: Subclasses

Translation:

MTDSMandatoryRegulatoryClass

Competency question: Which MTDS classes belong to prohibitory signs?

Method: DL Query

Protégé option: Subclasses

Translation:

MTDSProhibitoryRegulatoryClass

Competency question: Which MTDS classes belong to direction or indication signs?

Method: DL Query

Protégé option: Subclasses

Translation:

MTDSDirectionIndicationClass

Competency question: Are there MTDS classes that do not have an exact Vienna Convention equivalent?

Method: SPARQL

Translation:

```
SELECT ?mtdsClass
WHERE {
  ?mtdsClass rdf:type owl:Class .
  FILTER(CONTAINS(STR(?mtdsClass), "MTDS"))
  FILTER NOT EXISTS { ?mtdsClass owl:equivalentClass ?equiv . }
  FILTER NOT EXISTS { ?equiv owl:equivalentClass ?mtdsClass . }
}
ORDER BY ?mtdsClass
```

Note: Finds MTDS classes without explicit equivalentClass mappings.

Competency question: What MTDS dataset label is associated with a class?

Method: SPARQL

Translation:

```
SELECT ?class ?label
WHERE {
  ?class ?p ?label .
  FILTER(CONTAINS(STR(?p), "hasMTDSClassLabel"))
}
ORDER BY ?class
```

Note: Dataset labels are literals, so SPARQL is appropriate.

Competency question: What image filename contains a given MTDS annotation?

Method: SPARQL

Translation:

```
SELECT ?annotation ?image ?filename
WHERE {
  OPTIONAL { ?annotation ?rel ?image . FILTER(CONTAINS(STR(?rel), "Image")) }
  ?image ?p ?filename .
  FILTER(CONTAINS(STR(?p), "hasImageFilename"))
}
```

Note: Requires MTDS image/annotation individuals. If the ontology is schema-only, this may return no rows.

Competency question: What are the width and height of an MTDS image?

Method: SPARQL

Translation:

```
SELECT ?image ?width ?height
WHERE {
  OPTIONAL { ?image ?wProp ?width . FILTER(CONTAINS(STR(?wProp), "hasImageWidth")) }
  OPTIONAL { ?image ?hProp ?height . FILTER(CONTAINS(STR(?hProp), "hasImageHeight")) }
}
```

Note: Requires image individuals with data values. For schema validation, query the domains/ranges of hasImageWidth and hasImageHeight.

Competency question: What dataset split does an MTDS image belong to?

Method: SPARQL

Translation:

```
SELECT ?image ?split
WHERE {
  ?image ?p ?split .
  FILTER(CONTAINS(STR(?p), "hasDatasetSplit"))
}
```

Note: Requires image individuals with split values.

Competency question: What are the bounding-box coordinates of an MTDS annotation?

Method: SPARQL

Translation:

```
SELECT ?annotation ?xmin ?ymin ?xmax ?ymax
WHERE {
  OPTIONAL { ?annotation ?xminProp ?xmin . FILTER(CONTAINS(STR(?xminProp), "hasBoundingBoxXMin")) }
  OPTIONAL { ?annotation ?yminProp ?ymin . FILTER(CONTAINS(STR(?yminProp), "hasBoundingBoxYMin")) }
  OPTIONAL { ?annotation ?xmaxProp ?xmax . FILTER(CONTAINS(STR(?xmaxProp), "hasBoundingBoxXMax")) }
  OPTIONAL { ?annotation ?ymaxProp ?ymax . FILTER(CONTAINS(STR(?ymaxProp), "hasBoundingBoxYMax")) }
}
```

Note: Requires annotation individuals with bounding-box values.

Competency question: Are MTDS metadata properties clearly distinguished from road-sign semantic properties?

Method: SPARQL

Translation:

```
SELECT ?property ?domain ?range
WHERE {
  ?property rdf:type owl:DatatypeProperty .
  OPTIONAL { ?property rdfs:domain ?domain . }
  OPTIONAL { ?property rdfs:range ?range . }
  FILTER(
    CONTAINS(STR(?property), "MTDS") ||
    CONTAINS(STR(?property), "Image") ||
    CONTAINS(STR(?property), "Annotation") ||
    CONTAINS(STR(?property), "BoundingBox")
  )
}
ORDER BY ?property
```

Note: Retrieves MTDS metadata properties and their domains/ranges.

Recommended Queries to Actually Execute

Although every CQ is formalised above, the strongest DL queries are the class-hierarchy and semantic-restriction queries. Property metadata and dataset metadata should be evaluated with SPARQL.

DL Queries to run in Protégé:

RoadSign	[Direct subclasses]
DangerWarningSign	[Superclasses]
ProhibitorySign	[Superclasses]
MandatorySign	[Superclasses]
AdditionalPanel	[Subclasses]
RoadSign and (appliesToRoadUser some Pedestrian)	[Subclasses]
RoadSign and (appliesToVehicle some Bicycle)	[Subclasses]
RoadSign and (expressesProhibition some Prohibition)	[Subclasses]
RoadSign and (expressesObligation some Obligation)	[Subclasses]
RoadSign and (modifiesSign some RoadSign)	[Subclasses]
RoadSign and (hasShape some SignShape)	[Subclasses]
MTDSDataSetSignClass	[Subclasses]
MTDSDataSetSignClass and RoadSign	[Subclasses]

SPARQL should be used for equivalent-class mappings, functional properties, inverse properties, MTDS labels, image filenames, image dimensions, dataset split, bounding boxes, and metadata-property separation.